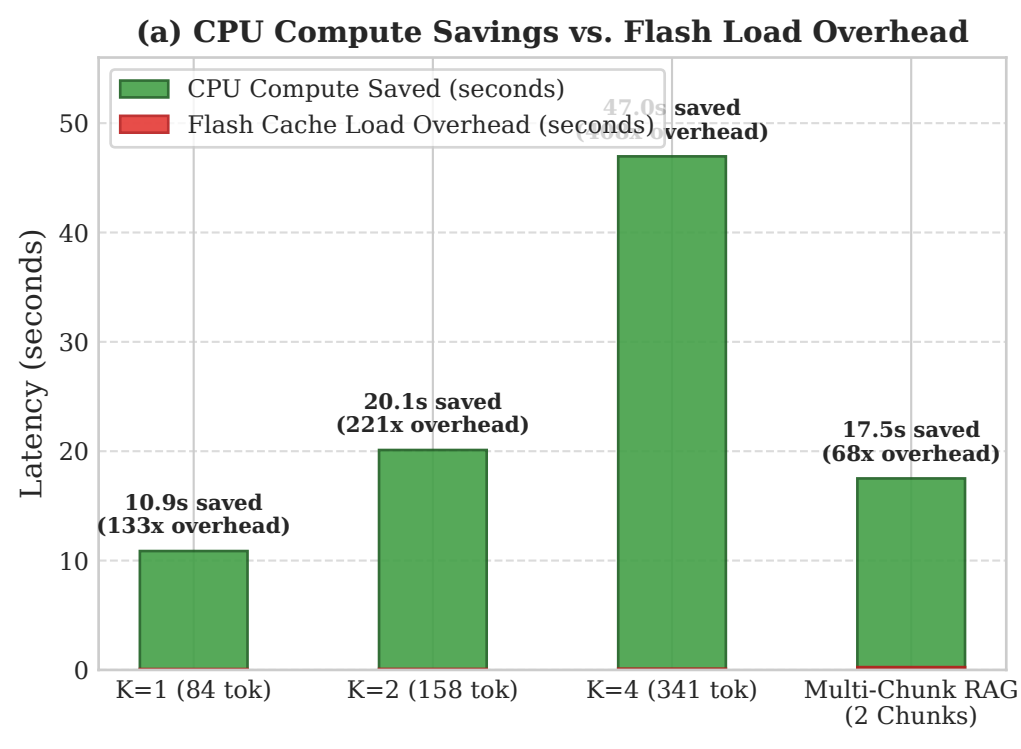
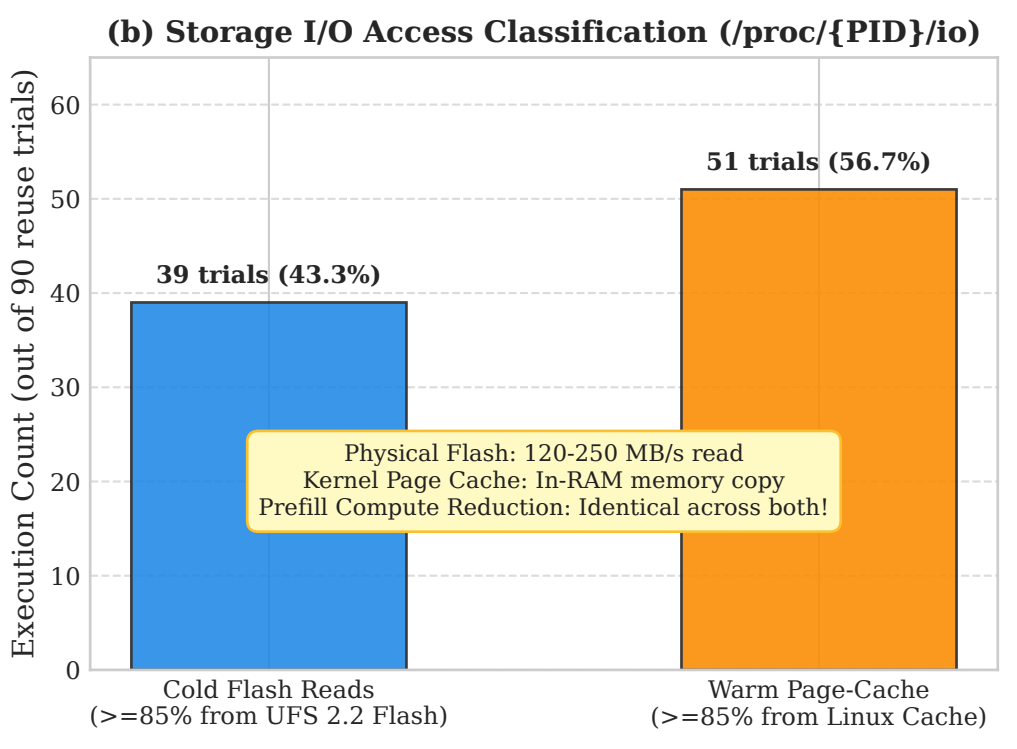

(a) CPU Compute Savings vs. Flash Load Overhead
CPU Compute Saved (seconds)
Flash Cache Load Overhead (seconds)
Latency (seconds)
0
10
20
30
40
50
10.9s saved
(133x overhead)
20.1s saved
(221x overhead)
47.0s saved
verhead)
17.5s saved
(68x overhead)
K=1 (84 tok)
K=2 (158 tok)
K=4 (341 tok)
Multi-Chunk RAG
(2 Chunks)


(b) Storage I/O Access Classification (/proc/{PID}/io)
Execution Count (out of 90 reuse trials)
0
10
20
30
40
50
60
39 trials (43.3%)
51 trials (56.7%)
Physical Flash: 120-250 MB/s read
Kernel Page Cache: In-RAM memory copy
Prefill Compute Reduction: Identical across both!
Cold Flash Reads
(>=85% from UFS 2.2 Flash)
Warm Page-Cache
(>=85% from Linux Cache)